

Phases for Project 1

Phase 1: Data Collection

- Collect images (JPG format) of COVID-19 devices
- At least 400 images
- Divide the data into train and test datasets
 - For example, 75% and 25% or 80% + 20%, etc.

Phase 2: Object Detection

- Boundary box labeling
- Investigate and use an appropriate tool
 - Examples: LabelMe or Roboflow
 - Labelling for any position of the object
 - [optional] Label as polygon with four points in the space
- Label the images – only the train set
- My recommendation is CVAT:
 - On the web site cvat.ai
 - Free download using cvat container

Phase 3: Training the object detection model

- Investigate and choose the AI model for object detection
 - e.g., YOLOv5, SSD, Faster R-CNN, YOLO8
- Goto <https://huggingface.co/> for more models